

E-Waste Component Classification with Agentic AI for Intelligent Recycling Decision Support

Aligned with SDG 12

(Predictive Analytics)

Presented by

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Abstract

Despite rising volumes of discarded electronics, toxic elements like lead, mercury, cadmium, yet lithium persist in landfills, threatening ecosystems along with human well-being. To respond, a staged artificial intelligence system emerges - designed not only to sort device parts precisely but also guide safer recycling pathways. Such an approach ties directly to UN SDG 12.4, aiming for responsible chemical and waste control worldwide.

A new approach combines deep learning, pattern detection without supervision, and forecasting danger levels within one streamlined process for smarter recycling decisions. Instead of training from scratch, existing networks like ResNet18 through ViT-B/16 adapted via transfer learning to sort 18 kinds of electronic waste parts with high precision. From top models, extracted features became inputs for techniques such as PCA, which shrinks data size, while MiniBatch K-Means and K-Medoids exposed groupings others might miss. Visual mapping followed, using t-SNE to reveal how different toxic elements relate in layout space. Beyond sorting, a neural network estimated threat intensity continuously - then grouped outcomes into three tiers: HIGH, MEDIUM, or LOW risk - based on physical traits and ecological impact measures.

Tests show ConvNeXt-Tiny alongside ViT-B/16 reached top results - 98.11% correct labels and a macro F1 of 0.9811 over 18 types of electronic waste. Because it scored 0.9846 on R^2 , the neural network model handled continuous hazard levels well; classification hit 95.56%. Lead detection matters - ahead of many inputs - as does knowing which components appear, since both steer risk forecasts sharply.

One key aspect of this study is building an integrated smart system for recycling. This setup brings together detailed part-by-part sorting along with automatic pattern detection, risk assessment, and AI-driven advice - all working inside one flow. Most current tools limit themselves to broad trash grouping. By contrast, the method introduced here targets dangers tied to individual components while offering practical directions for eco-friendly handling. What sets it apart is how precisely it handles materials in actual use cases.

Because it uses artificial intelligence to automate processes, the system supports safer management and smarter sorting of electronic waste. This approach encourages recycling that recognizes hazards while ensuring disposal methods protect ecosystems. Such capabilities line up closely with targets under Sustainable Development Goal 12.4. Over time, the structure allows room for newer tools like visual pattern models and self-guided reasoning agents. These additions could help build recycling networks that operate independently yet remain transparent in their decisions.

Index Terms: e-waste classification, transfer learning, ResNet, EfficientNet, Vision Transformer, hazard, ANN, k-means clustering, t-SNE, SDG 12.4, sustainable recycling

I. Introduction

One of the waste streams with the fastest rate of growth worldwide is electronic waste, also known as e-waste. According to the Global E-Waste Monitor, the generation of e-waste is rising beyond 62 million metric tonnes in 2022. However, only 25% of it is being recycled. One of the key challenges in automated recycling is identifying individual internal components of discarded electronic devices such as batteries,

printed circuit boards, microchips, capacitors, resistors, and semiconductors. These components are known to contain hazardous materials such as lead, mercury, cadmium, and brominated flame retardants. The process of manual sorting of waste in recycling plants is highly labor-intensive and error-prone. Currently, existing automated waste classification systems are only capable of identifying waste products in general terms, such as "mobile phone" or "laptop." This is an obstacle in achieving sustainable development goal 12.4, which demands the safe management of chemicals and waste throughout their whole life cycle.

This research introduces a predictive analytics-based solution to overcome these challenges in waste component classification and hazard assessment. A multi-stage predictive analytics-based solution is proposed. In this solution, this research is organized in such a manner: (i) deep learning-based computer vision models ResNet18, ResNet50, EfficientNet-B0, ConvNeXt-Tiny, Swin-Tiny and ViT-B16 via transfer learning for component-level 18-class classification; (ii) unsupervised embedding-based clustering for hazard assessment pattern discovery; and (iii) an artificial neural network for hazard prediction. In this research, we are also introducing an analysis of the performance of seven classifiers. Therefore, this research is also fulfilling another requirement of experimentation and benchmarking of multiple classifiers.

The rest of this research is organized in such a manner that Section II deals with the literature review, Section III discusses research gaps, Section IV discusses proposed methodology, Section V discusses datasets, Section VI discusses implementation, Section VII discusses experimental results and analysis, and Section VIII deals with conclusion.

II. Literature Survey

The rapid growth of e-waste generation has encouraged significant research around automated classification, material identification, and smart recycling techniques. This section discusses some of the important developments in the following five thematic areas, which are of particular interest to the current work.

A. Transfer Learning-Based Convolutional Neural Network Architectures for E-Waste Component Classification

When handling automated waste separation, Convolutional Neural Networks often come first due to their strong grasp of image patterns. Research into boosting these models has taken many paths, each aiming at better performance without added complexity.

Kukade et al. began with ImageNet's pre-trained weights, then shifted to transfer learning using CNNs like ResNet and EfficientNet for classifying trash photos. Over 90 percent accuracy showed up in category detection thanks to their technique, while time spent on training dropped fast. High results stayed within reach despite limited labeled examples because of this path (Kukade et al., 2024).

Out of left field, Oise alongside Konyeha rolled out a Sequential Neural Network meant for sifting through electronic trash, trained using information pulled from Kaggle. Not flashy by any means, yet it hit close to 91% correctness. That kind of result hints at smaller setups handling real work when heavy-duty computing isn't an option (Oise and Konyeha, 2024).

Ahmad and colleagues tested a mix of CNN setups - VGG16, ResNet50, alongside InceptionV3 - inside the TrashNet system. Instead of using one model alone, they combined predictions then passed them to an SVM. Accuracy reached 96.2%, outdoing each individual network. The findings revealed teamwork beat solo efforts when performance lagged in separate runs. (Ahmad et al., 2020).

Starting with image-based sorting, Mulim and team applied EfficientNet-B0 to categorize waste across six types. Reaching nearly 93.4 percent precision, the system stayed lightweight in processing demands. Because of its modest resource needs, operation during live workflows remains feasible. Though compact, performance does not drop sharply under practical conditions (Mulim et al., 2021).

Starting strong, Yang et al. developed GarbageNet, a convolutional neural network integrating attention mechanisms. This system underwent testing using a collection of 15,000 pictures sorted into twelve classes. Performance results showed an accuracy rate reaching 97.1 percent. Despite shifts in surroundings during tests, it held up well, proving resilience in diverse settings. (Yang et al., 2021).

Starting with work by Vo et al., a method based on VGG16 and InceptionV3 was tested on a tailored collection of images, reaching 94.3% correct predictions. Performance jumped when earlier trained networks were applied - especially useful where data is limited (Vo et al., 2019).

Not limited to software alone, Tanveer and colleagues introduced a deep learning model for intelligent waste sorting that combines computer vision and IoT systems to support live tracking and categorization. What stands out is how convolutional neural network methods can function effectively outside lab settings (Tanveer et al., 2023).

Looking at those earlier studies, it becomes clear that CNN-driven methods using transfer learning perform well when sorting waste. Still, the majority of prior work zeroes in on identifying entire products instead of breaking them down into individual parts - yet such detail is necessary when assessing dangers tied to electronic waste.

B. Transformer-Based Visual Recognition Architectures for E-Waste Component Classification

Though CNNs perform strongly in sorting images, their narrow field can limit broader context awareness. Because of this gap, scientists have turned toward Vision Transformers. These models capture connections across distant regions more effectively. Their design allows a wider grasp of spatial relationships within visuals.

Surprisingly high results emerged when Huang et al. turned to Vision Transformers for sorting waste - over 96% correct classifications appeared across standard test sets. Rather than relying on conventional convolutional networks, this approach gained strength through self-attention, pulling together distant image features into meaningful patterns. What stood out was how well it handled broad spatial context, a task where older models often struggled (Huang et al., 2021).

A different approach emerged through Yang and colleagues' work, introducing WMC-CNN - an adjusted transformer paired with convolutional traits - achieving 94.27% precision while using less computational load (Yang et al., 2022).

Starting with a dual-path design, EWasteNet by Islam et al. uses transformers to merge broad patterns with fine details. This method maintains strong performance even when trained on limited samples. Accuracy stays high due to balanced feature integration across streams (Islam et al., 2023).

Starting with a fresh approach, Chen et al. introduced LPCNN to classify PCBs by leveraging the DeepPCB dataset. Improved accuracy emerged through more effective capture of overall patterns. Global context was interpreted with higher clarity, leading to refined results (Chen et al., 2022).

Recently, attention turned to speeding up such models while boosting efficiency. Bai and colleagues proved effective training methods let Transformers stay strong performers without massive data (Bai et al.). Building on that idea, Zhou's team introduced a mixed design. Features first get captured through convolutions - then linked via self-attention, sharpening detection of subtle patterns across inputs (Zhou et al., 2024).

Despite promising results in identifying broad patterns within waste streams, transformer-driven approaches face hurdles outside lab settings. Heavy processing demands slow down field adoption significantly. Their need for vast amounts of labeled data restricts integration into current sorting infrastructure. Real systems often lack the scale required to unlock their full capability. Performance gains come at a cost that many facilities cannot support.

C. Component-Level E-Waste Object Detection and Multi-Object Visual Recognition

Looking closer at single parts inside gadgets marks a shift from old methods that sorted whole products. Progress like this opens paths toward finer recovery in reuse systems.

Starting with YOLOv7, Chowdhury et al. focused on spotting e-waste parts - like PCBs, metals, and plastics - as they appear. Their method achieved a mAP of 0.96 using a specially built collection of images, showing strong accuracy. Real-time speed was confirmed through testing under live conditions (Chowdhury et al., 2024).

Starting with object detection, Nguyen et al. applied YOLOv8 alongside a valuation component to spot e-waste parts while gauging their monetary worth. Instead of just labeling items, the approach delivered cost-related data useful in recycling workflows (Nguyen et al., 2024).

Rojas and team applied Mask R-CNN alongside YOLACT++ to isolate inner parts of dismantled gadgets. With an IoU reaching 0.79, the top-performing system delivered precise identification at the part level (Rojas et al., 2022).

Working on tiny parts recognition, Rajeev et al. introduced a refined YOLO structure that adjusts across scales using focused attention layers. Meanwhile, Xu et al.'s version improved detection accuracy by integrating similar scale-aware mechanisms into their model design (Xu et al., 2025).

Though tools such as YOLO and Mask R-CNN perform well at spotting individual parts, their role often stops short of evaluating risks. Instead of guiding choices, they concentrate mostly on finding and pinpointing objects within images.

D. Sensor Fusion and Hyperspectral Imaging Techniques for Advanced E-Waste Material Analysis

Traditional RGB-based vision systems struggle to distinguish visually similar materials, leading to the adoption of sensor fusion techniques.

Hussein et al. used hyperspectral imaging combined with deep learning models to classify e-waste materials. The approach reached a classification accuracy above 90% by capturing spectral signatures beyond visible light (Hussein et al., 2025).

Picon et al. developed a hyperspectral dataset for Waste Electrical and Electronic Equipment (WEEE) classification. They demonstrated enhanced material identification compared to RGB-based systems (Picon et al., 2024).

Recent advancements include multimodal approaches that combine various data sources. Chen et al. proposed a multimodal deep learning framework integrating both visual and spectral data, achieving improved robustness in material classification tasks (Chen et al., 2022).

However, these systems need expensive hardware and complex calibration, constraining their practical deployment in recycling facilities.

Sensor fusion and hyperspectral imaging improve material-level classification accuracy significantly. However, due to high cost and complexity, RGB-based deep learning models remain more practical for scalable real-world applications.

E. AI for Recycling Optimisation and Decision Support

Recent research has expanded from classification to include intelligent decision-making systems.

Deep learning helped streamline logistics in e-waste collection, thanks to work by Nowakowski and Pamula. Smarter routes emerged - not through guesswork, but data patterns spotted automatically. Efficiency improved because systems learned from past runs. Planning shifted subtly over time, adapting without constant human input. The method reduced delays simply by anticipating needs ahead of time (Nowakowski and Pamula, 2020).

Using artificial intelligence, Madhav and team improved trash pickup routes. This led to less fuel use along with lower running expenses. Their method linked smart computing to real-world logistics. Efficiency rose without raising complexity. Savings emerged not just in energy but also in time spent. The approach shifted how cities manage garbage transport. Results showed steady gains across different urban layouts (Madhav et al., 2022).

By pairing cost analysis with material identification, Nguyen and colleagues improved choices in waste recovery processes (Nguyen et al., 2024).

A different approach emerged through Talwar et al., who introduced a deep learning model aimed at identifying recyclable materials automatically. Efficiency in sorting processes was central to their work, driven by neural network techniques applied without manual oversight (Talwar et al., 2026).

Recently, Singh and Kumar introduced a decision support tool powered by artificial intelligence aimed at advancing e-waste sustainability. Their approach highlights blending environmental insights into analytical models through machine learning techniques. Focus shifts toward smarter recycling processes when such systems guide operational choices. Integration becomes central when data shapes ecological impact reduction. Outcomes improve not solely due to technology, but because context-aware analysis informs decisions. Emphasis lands on how intelligent processing supports long-term resource recovery goals (Singh and Kumar, 2024).

Still, the majority of such frameworks handle classification, optimization, and decisions as isolated stages instead of combining them within one cohesive flow.

Though artificial intelligence aids recycling operations and choices, few complete systems join sorting, evaluation, with risk forecasting in one smart platform.

III. Research Gaps and Motivation

A systematic review of the literature reveals the following significant gaps motivating the present research:

A. Lack of Fine-Grained Classification of Waste Type into Individual Components

Most of the prior work classifies waste into broad product categories (e.g., mobile phone, laptop, television) rather than identifying discrete internal components such as capacitors, PCBs, microchips, or semiconductor diodes. Fine-grained component-level identification is fundamental to hazard-specific recycling decisions. The proposed system addresses this through classification across 18 component-level categories.

B. Lack of Integrated Hazard Assessment

Current automated systems do not often go beyond the identification of the components to assess the level of environmental or health hazards. Components such as batteries, PCBs, and semiconductors contain hazardous materials such as lead, lithium, mercury, and cadmium, but the automated systems do not assess the level of hazard and make recommendations regarding their disposal. The ANN-based hazard prediction module of the proposed system addresses this problem.

C. Lack of Unified Multi-Stage AI Pipelines

Previous studies primarily focus on individual problems, such as classification, detection, clustering, or logistics, but fail to incorporate them into a comprehensive decision-support system. The proposed pipeline combines supervised component classification, unsupervised hazard pattern clustering, and regression-based hazard scoring, making it applicable to the recycling facility setting.

D. Limited Application of Vision Transformers to E-Waste

Although Vision Transformer have shown better performance in fine-grained recognition with global context, their adoption in e-waste component classification is limited. The visual similarity of components such as microchips, integrated circuits, and semiconductor diodes makes this domain an appropriate candidate for the adoption of the attention mechanism of the transformer. This paper proposes a baseline for CNNs and outlines the way to incorporate CNNs.

E. Performance Degradation on Visually Similar and Minority Classes

The high accuracy in the headlines of previous studies hides the severe degradation in the performance of the model on rare and visually ambiguous classes. The current study offers empirical analysis of the per-class performance of all 18 component categories.

IV. Proposed Methodology

The proposed system comprises three interconnected analytical stages: image-based component classification, unsupervised clustering for pattern discovery, and ANN-based hazard prediction. Figures 1 and 2 depict the system-level architecture and detailed proposed pipeline, respectively.

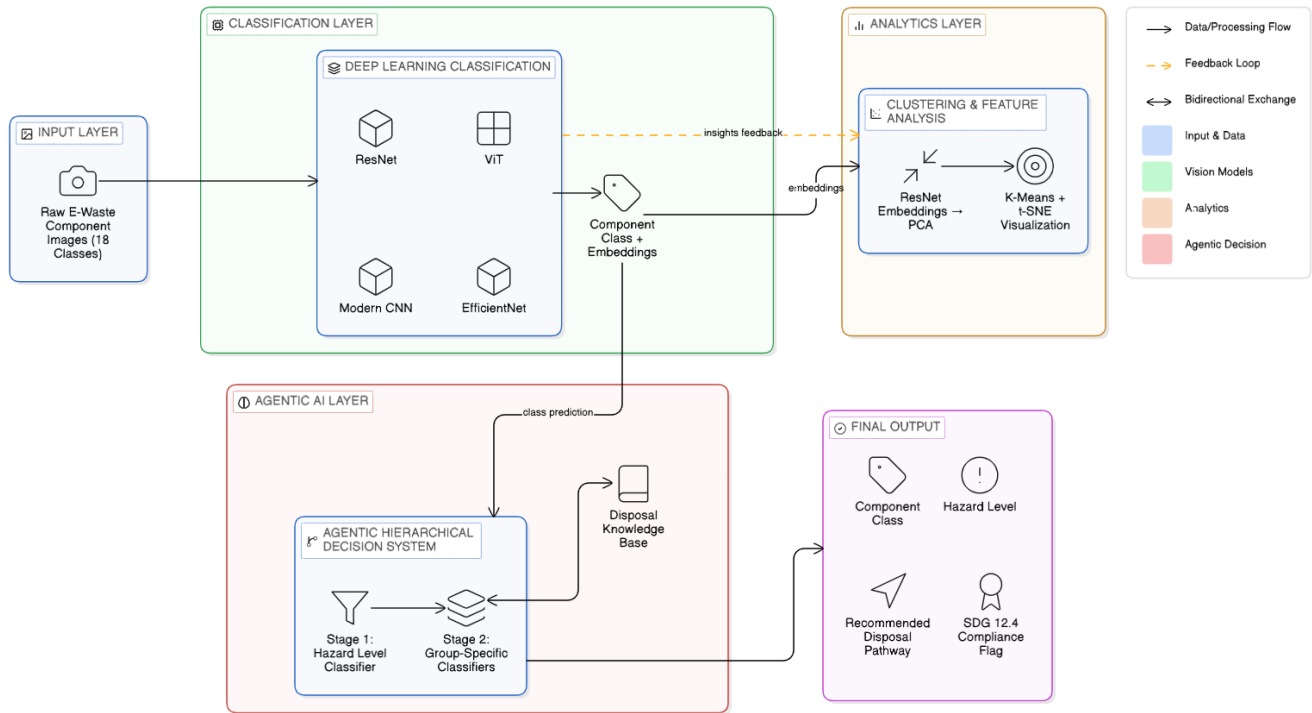


Figure 1. System architecture diagram of the proposed intelligent e-waste component classification and hazard-aware disposal framework

Figure 1. displays an advanced setup for handling electronic waste, split into four key sections: Input, Sorting, Analysis, and Smart Decision layers. Starting off, unprocessed images of discarded electronics enter the system. Instead of being used right away, these visuals go through powerful pattern-spotting networks like ResNet, EfficientNet, and ViT to label parts and pull out identifying traits. Once pulled apart digitally, those digital fingerprints get explored using dimension-reduction tools, group-finding math, and layout mapping techniques to uncover connections not obvious at first glance. After sorting, details about each part shift toward the reasoning engine, where danger scores form based on stored rules about safe handling methods. Out comes a full profile per item: type identified, risk grade assigned, best route chosen, plus alignment checked against global target 12.4 all enabling smarter choices in managing old gadgets.

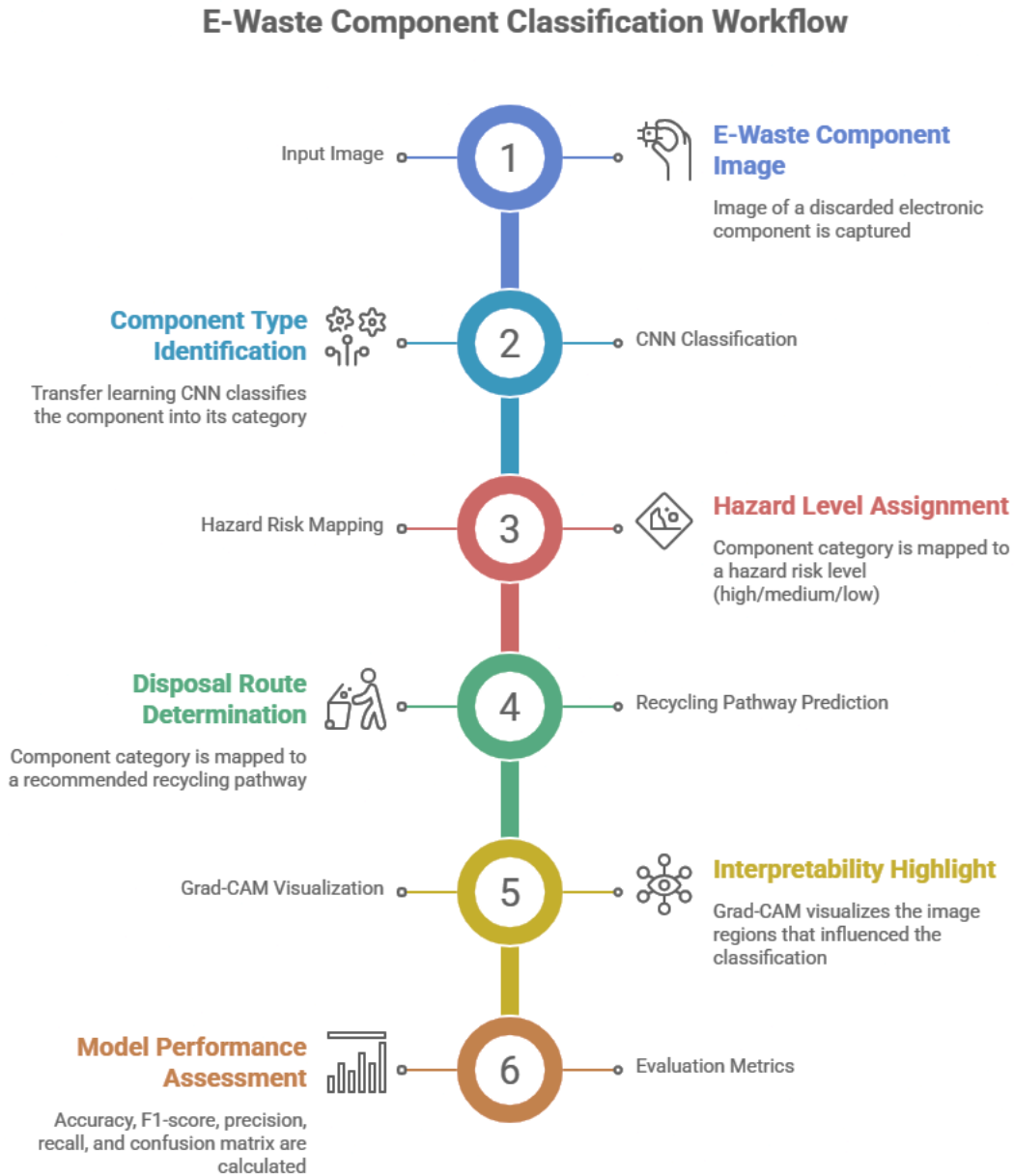


Figure 2. Workflow diagram of the proposed intelligent e-waste component classification and hazard-aware disposal framework.

Figure 2 reveals how the new e-waste sorting method works. First comes snapping a photo of an electronic part, after that a convolutional neural network labels what kind it is. Once labeled, each piece gets tagged with its risk level, so it can be routed toward proper reuse or safe discarding. To see which areas shaped the model's decision, heatmaps from Grad CAM point out key zones in the image. On evaluation, results rely on standard indicators like correctness rate, exactness, completeness, balance score, along with mismatch charts to measure outcomes.

The framework operates through the following four sequential layers, which are detailed in the subsequent subsections:

A. Perception Layer: Deep Learning-Based E-Waste Component Image Classification

The perception layer implements visual component recognition as a supervised single-label classification task across 18 e-waste component categories. The input dataset comprises 27,560 images organized into three splits: 23,960 training, 1,800 validation, and 1,800 test samples, with each image resized and square-padded to 224×224 pixels using bicubic interpolation.

Deep Learning Baseline

Transfer learning was therefore employed using seven state-of-the-art deep architectures, all initialized with ImageNet pre-trained weights. The evaluated models were:

- **ResNet18** – a lightweight residual network offering fast convergence
- **ResNet50** – a deeper residual network providing richer hierarchical feature representations
- **EfficientNet-B0** – a parameter-efficient model based on compound scaling
- **EfficientNet-B3** – an enhanced EfficientNet variant with improved capacity
- **ConvNeXt-Tiny** – a modern convolutional architecture inspired by transformer design principles
- **Swin-Tiny** – a hierarchical vision transformer using shifted windows for multi-scale feature modelling
- **ViT-B/16** – a transformer-based model that captures global dependencies through self-attention

All models underwent a two-stage training protocol: an initial 5 epochs with a frozen backbone and trainable classification head, followed by 35 epochs of full-network fine-tuning. Classification heads were architecture-specific (ResNet models: 512→256 bottleneck with batch normalization and dropout; ViT - B/16: 512→256 projection).

To improve the model performance and handle class imbalance, different data augmentation techniques were used during training. They included random horizontal flipping, small image rotations (upto 15 deg), random cropping, color adjustments like brightness and contrast changes, grayscale conversion and random erasing. We also added a random sampler to the data loader so that all classes were represented more evenly during training.

For optimisation we used the AdamW optimiser (learning rate 1×10^{-4} , weight decay 1×10^{-4}). We trained the model using cross-entropy loss and we also added label smoothing with a value of 0.05 to reduce overconfidence and improve generalization. Mixed-precision training with automatic loss scaling was enabled on CUDA devices. Early stopping (patience=8 epochs) and learning rate reduction provided regularization. Data augmentation during training included letterbox-based square padding, random horizontal and vertical flips, rotation ($\pm 10^\circ$), colour jitter, random auto contrast, and random erasing. Class imbalance was addressed via weighted sampling during training. ConvNeXt-Tiny and ViT - B/16 achieved the highest test accuracies among all architectures, validating modern vision transformer and hierarchical convolution designs for this domain.

Among the seven deep architectures, **ConvNeXt Tiny** and ViT - B/16 demonstrated the highest test accuracy and the strongest overall performance. ConvNeXt was therefore selected as the primary feature extractor for generating embeddings used in subsequent clustering and traditional machine learning stages.

Traditional Machine Learning Baseline

A hierarchical Support Vector Machine (SVM) pipeline was implemented to benchmark classical approaches. A two-stage architecture applied RBF-kernel SVC ($C=10.0$, $\gamma='scale'$) first to classify hazard level (HIGH/MEDIUM/LOW) globally, then per-hazard second-stage classifiers refined fine-grained component predictions. Feature inputs were 256-dimensional PCA-reduced embeddings extracted from the penultimate layers of the best deep model (ResNet50) after standardization. This comparison showed that supervised deep learning models performed much better than traditional kernel-based methods for this visual recognition task. The deep learning approaches were able to learn more meaningful image features, which helped improve the overall classification performance.

In summary, the perception layer converts raw e-waste images into semantically rich feature representations that can be used for accurate classification. We compared different CNN and transformer-based models with a classical baseline model to properly evaluate their performance. From the results, we found that the deep learning models provided a much more robust and reliable foundation for the overall intelligent recycling pipeline.

B. Analytics Layer: Unsupervised Clustering and Feature Embedding Analysis for E-Waste Components

Because the analytics layer uses unsupervised clustering, it finds inherent visual clusters among various e-waste parts without relying on initial category tags. Hidden trends emerged - shared shapes, surface qualities, or makeup - that were not obvious at first glance. These insights support improved sorting methods, along with more informed choices in recycling processes.

One layer before the output in ConvNeXt Tiny delivered stronger results than other models, so that is where we pulled 512-element feature vectors. To help speed up grouping steps while simplifying visual inspection, these features went through standardization followed by compression via PCA.

During deployment, one method was set beside another for contrast. Each algorithm faced similar conditions yet followed distinct paths. Their performance emerged through side-by-side observation rather than isolated tests. Differences became clear only when results were matched step by step. This way of working revealed strengths not obvious at first glance

K-Means Clustering: Applied as a baseline to partition the **27,560 samples**.

A different approach was taken by using a sampled version of PAM within a **K-Medoids** framework - this way, centers emerged directly from real data instances rather than averages. Instead of abstract central points, each cluster now holds an observed entry as its core anchor. Through CLARA-inspired sampling, computational load decreased while keeping results grounded in actual observations. The method shifts focus from theoretical means to tangible examples present in the original collection.

Three clusters were chosen, matching the detailed hazard categories. This choice followed analysis through several measures of cluster quality - patterns seen in the Elbow Method, values from the Silhouette Score, results of the Davies–Bouldin Index, along with findings from the Calinski–Harabasz Score. Though not perfect, each metric pointed toward three as a sensible count.

Looking at patterns visually, t-SNE mapped complex feature data down to two dimensions. Though shapes appear in the plots, groupings blur together more than they stand apart. Instead of clean divisions matching labels like HIGH, MEDIUM, or LOW, boundaries stay fuzzy. What emerges does not align neatly with expected risk classes.

Despite weak separation, the unsupervised analysis confirmed how well the feature embeddings performed. Because of this, it offered actionable insights into improving actual e-waste sorting and recycling workflows.

C. Analytics Layer: ANN-Based Hazard Score Prediction and Risk-Level Assessment for E-Waste Components

The hazard prediction layer uses a feedforward Artificial Neural Network (ANN) to classify e-waste components into hazard categories based on structured tabular data. The dataset contains **27,560 samples** covering 18 different component types, including **Batteries, PCBs, Microchip-ICs, mobile devices, and other electronic parts**.

The input dataset includes 18 classes of e-waste components mapped to their respective hazard levels: **HIGH, MEDIUM, and LOW**. The classification logic is specifically made to identify components containing hazardous materials such as **lead, mercury, and cadmium**—to ensure compliance with **SDG 12.4** guidelines for environmentally sound waste management. During training, we processed the categorical variables properly so the model could learn the patterns more effectively and make accurate hazard-level predictions.

To ensure the highest possible reliability, we conducted a detailed optimization study by comparing different backpropagation algorithms, and Adam gave the best overall performance and achieved a training accuracy of 98.08%.

These improvements also showed a positive translation to real-world safety applications. The final model achieved an overall test accuracy of 98.56%, and a precision score of 0.99 for detecting high-hazard e-waste materials. This shows that the model is very effective at identifying components that may require careful handling and disposal.

D. Agentic AI-Based Decision Support and Hazard-Aware Disposal Recommendation Module

V. Dataset Description

Built from the ground up, this work delivers a detailed collection of e-waste components, shaped to meet gaps in available information about internal electronics. Where most prior sets zoom out to entire devices - like laptops or phones - a closer lens emerges here, targeting individual parts instead. Shaped by necessity, the new resource supports precise recognition and risk evaluation at a level previous efforts rarely reach.

A. Major Contribution: Novel 18-Class E-Waste Dataset

A total of **27,560 labeled images** form this collection, covering **18** separate categories. Because it focuses on internal parts - such as **heat-sinks, transistors, and passive** elements - the data supports detailed analysis of electronic waste. Where earlier studies lacked such detail, this work fills a clear gap. Available now on Kaggle, the release encourages consistent results in research tied to responsible recycling practices. With alignment to Target 12.4 of the Sustainable Development Goals, accessibility becomes part of its purpose.

B. Methodology: Systematic Curation and Selection

The dataset was developed through a rigorous multi-stage curation process:

- From online sources, images were collected methodically - pulling data from scattered public databases dedicated to discarded electronics. Various digital equipment guides contributed further material over time. Visual records hosted on internet platforms formed another key source throughout the gathering phase.
- Choosing by hand, each image picked carefully so objects stood apart, easy to recognize during learning. Precision mattered most; only clean examples made the cut.
- Starting with safety concerns, the 18 groups include items people use daily - such as refrigerators and washing machines - as well as hidden components like microchips, circuit boards, and batteries. These internal pieces often carry harmful substances including lead or mercury. Because of their risk potential, they share space within the classification system. Though different in function, each plays a role in product toxicity. From household appliances to tiny electronic elements, grouping them highlights shared environmental dangers. This structure allows clearer tracking across diverse but related objects.

C. Dataset Description and Distribution

Splitting the data carefully helps deep learning models train more effectively, especially when some categories appear less often. Distribution balances across groups reduce distortion from uncommon types. One portion serves training purposes alone, containing 23,960 instances. Another segment, set aside for validation, includes 1,800 samples. The last group, reserved strictly for testing, also holds 1,800 entries.

TABLE I. Class-Wise Distribution of the E-Waste Component Image Dataset

Class	Train	Val	Test	Total
Air-Conditioner	2114	100	100	2314
Battery	1160	100	100	1360
Keyboard	1259	100	100	1459
Laptop	1415	100	100	1615
Microchip-IC	3556	100	100	3756
Microwave	1113	100	100	1313
Mobile	839	100	100	1039
Mouse	800	100	100	1000
PCB	841	100	100	1041
Passive-Component	2692	100	100	2892
Printer	1661	100	100	1861
Refrigerator	1098	100	100	1298
Resistor	800	100	100	1000
Television	800	100	100	1000
Washing Machine	799	100	100	999

heat-sink	800	100	100	1000
light bulbs	1413	100	100	1613
transistor	800	100	100	1000
TOTAL	23960	1800	1800	27560

A. Tabular Dataset for Hazard Score Prediction and Environmental Risk Assessment

One way to start building the dataset involved gathering details from the Global E-waste Monitor along with guidance under the Basel Convention, ending up with 2,700 organized entries. Features include encoded types of components, how old each device is, its mass, whether it contains lithium, lead, mercury, or cadmium shown through yes-or-no flags, if recycling applies, what materials make it up, physical state, coded ways of discarding, plus a regional risk index tied to location. The goal? Predicting a danger rating that runs on a scale from zero to one hundred. With both sets available, testing can happen without internet access while still meeting data protection rules.

VI. Implementation

A software solution for e-waste analysis runs on **Python version 3.11.9**. Built around **PyTorch**, it handles neural network computations, whereas **scikit-learn** manages data preparation and group discovery tasks. Since many recycling setups lack advanced hardware, operation without a GPU remains fully supported. Still, when available, CUDA enables faster model training through reduced numerical precision. Despite lean configurations, performance gains emerge during intensive phases.

A. Computer Vision and Perception Layer

Among the tested models are **ResNet18**, **ResNet50**, **EfficientNet-B0**, **EfficientNet-B3**, **ConvNeXt-Tiny**, **Swin-Tiny**, and **ViT-B/16** - each evaluated through a benchmark system built into the perception component. While some rely on convolutional hierarchies, others use shifted windows or attention mechanisms; performance comparisons emerge across these seven modern designs.

- Before processing, image inputs undergo normalization based on ImageNet data, followed by resizing through bicubic interpolation to match required dimensions.
- Training adjustments happen through AdamW, where model weights shift gradually under a specific learning rate alongside controlled weight decay.
- Training relies on **Cross-Entropy Loss** paired with 0.05 label smoothing - this reduces excessive confidence in predictions while boosting performance across look-alike categories. Though subtle, the adjustment shifts how probabilities are assigned during learning, nudging the model toward more balanced decisions when distinctions blur between near-identical inputs.
- A short warm-up phase begins the process - five epochs in total - with the main network structure locked. Then comes extended adjustment across thirty-five rounds, where all layers adapt together.

B. Analytical Clustering and Feature Embedding

Patterns emerge through unsupervised learning as the system pulls rich feature representations from the second-to-last stage of **ConvNeXt-Tiny** - the top model. Though hidden at first, these structures become clear via high-dimensional embedding extraction during analysis.

- Feature vectors with 512 dimensions get scaled first, then compressed using PCA - keeping 95 percent of variation intact. While standardizing comes before, the main drop in size follows after.
- Starting from cluster setup, a comparison unfolds - **MiniBatch K-Means** versus CLARA-driven K-Medoids - mapped across risk levels labeled High, Medium, and Low. Though differing in method, both follow tier alignment closely. One relies on mean centers, the other on representative objects. Where one processes batches quickly, the second resists outliers more firmly. Each assigns points into groups matching prior hazard definitions. Despite distinct mechanics, outcomes serve the same structural purpose. Not every detail matches, yet grouping logic stays consistent throughout.
- Spreading out complex data patterns into a flat view happens through **t-SNE**. Thirty values shape the local balance of points nearby. A thousand steps guide how positions shift during layout. The final image shows clusters where meanings diverge clearly across space.

C. ANN-Based Hazard Regression and Classification

A different kind of prediction method uses a single network to estimate several danger levels at once.

- A layered network takes twelve numerical inputs - such as lead, mercury, cadmium levels and total weight - from structured data. This model moves information forward through multiple stages without loops or feedback paths.
- With attention to detail, the model uses **Huber Loss** to handle outliers during optimization. It relies on Adam as its update rule, balancing speed and stability. Accuracy climbs steadily until it hits 98.08% on training data. Precision emerges not from complexity but consistent refinement.
- Through post-training **gradient sensitivity analysis**, normalized mean absolute gradients are computed - revealing that "Component Class" strongly influences predictions. Environmental risk turns out especially responsive to changes in "Lead Presence." What stands out is how these two features dominate model behavior after training.

D. Agentic AI Decision Support Layer

Beginning with prediction data, the system feeds classified components and risk levels into an adaptive **AI** agent through the **Grok** interface. Instead of fixed rules, it uses generated logic to interpret results and form clear guidance on waste handling. From classification outcomes, actionable steps emerge - like designated deconstruction methods or secure containment options. **Compliance links appear next: each suggestion ties directly to targets under SDG 12.4.** Output transforms numbers into practical directives, steering disposal toward regulated protocols.

VII. Experimental Results and Analysis

A. Comparative Performance Evaluation of Deep Learning Architectures for E-Waste Component Classification

Looking at Table II, results show how different deep learning setups performed when classifying unseen data. Topping the list was **ConvNeXt Tiny**, reaching an accuracy of **98.11%** along with a matching macro F1-score of **0.9811**. Despite close competition, it outdid well-known models like ResNet50 - recorded at 97.89% and 0.9789 - and also beat Swin Tiny at 97.67%, 0.9767, plus ResNet18 trailing at 96.78%. Even though VIT-B/16 tied in both metrics exactly, hitting 98.11% on accuracy and score alike, one key difference stood clear. What made **ConvNeXt Tiny** stand apart wasn't just precision but how fast it trained, proving far more efficient overall.

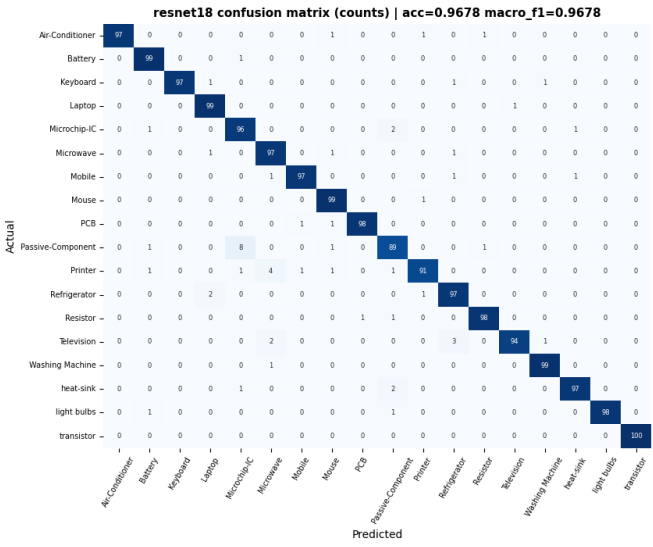
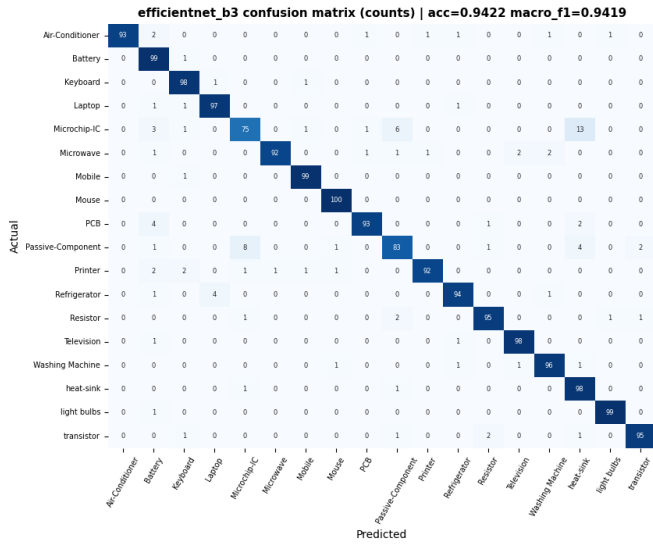
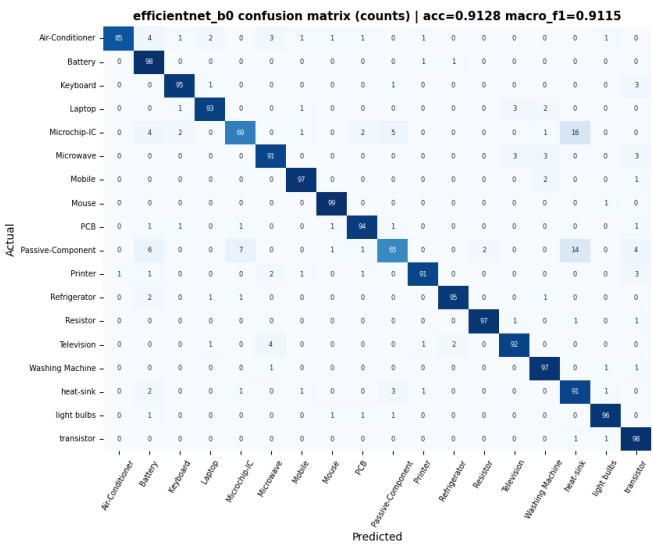
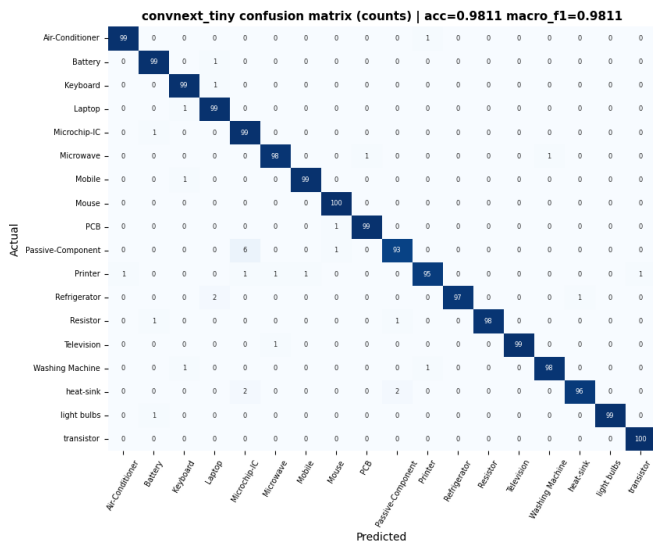
The findings point toward ConvNeXt's updated convolution strategy - blending features inspired by Vision Transformers - as best capturing the intricate visual patterns within the 18-category **e-waste** classification. While models built on transformer logic performed well, it was the refined spatial assumptions in **ConvNeXt Tiny** that delivered superior stability across this data. For what followed, including danger evaluation, tiered sorting, and generating dense feature vectors for grouping behavior, this model became the chosen base structure.

TABLE II. Comparative Test-Set Performance of Transfer Learning-Based Deep Learning Architectures for E-Waste Component Classification

Metric	ResNet18	ResNet50	Efficient-B0	Efficient-B3	ConvNeXt Tiny	Swin Tiny	VIT-B16
Accuracy	96.78%	97.89%	91.28%	94.22%	98.11%	97.67%	98.11%
Macro F1	0.9678	0.9789	0.9115	0.9419	0.9811	0.9767	0.9811
Macro Precision	0.9684	0.9791	0.9159	0.9441	0.9815	0.9773	0.9818
Macro Recall	0.9678	0.9789	0.9128	0.9422	0.9811	0.9767	0.9811
Test Loss	0.4950	0.4537	1.0918	0.5773	0.4332	0.4506	0.4312

Looking at Figure 3, normalized confusion matrices appear for several tested models: **ConvNeXt Tiny**, **VIT-B/16**, **ResNet50**, and **Swin Tiny**. Among them, ConvNeXt Tiny along with VIT-B/16 shows stronger concentration on the diagonal. This pattern suggests these two handle class separation more effectively than others do.

Most models handle items like **Mouse**, **Transistor**, **Battery**, and **Laptop** with high accuracy, regardless of design differences. Classes including **Keyboard** and **Light Bulbs** show stable results too, across test runs. Yet some tiny electronics remain tricky to tell apart. For instance, **Passive-Components** often get labeled as **Microchip-ICs** by mistake. Their shape plays a role - both tend to be rectangular. Surface appearance adds to the mix: shiny metal finishes appear in both groups. Size further blurs boundaries since they occupy similar space in images.



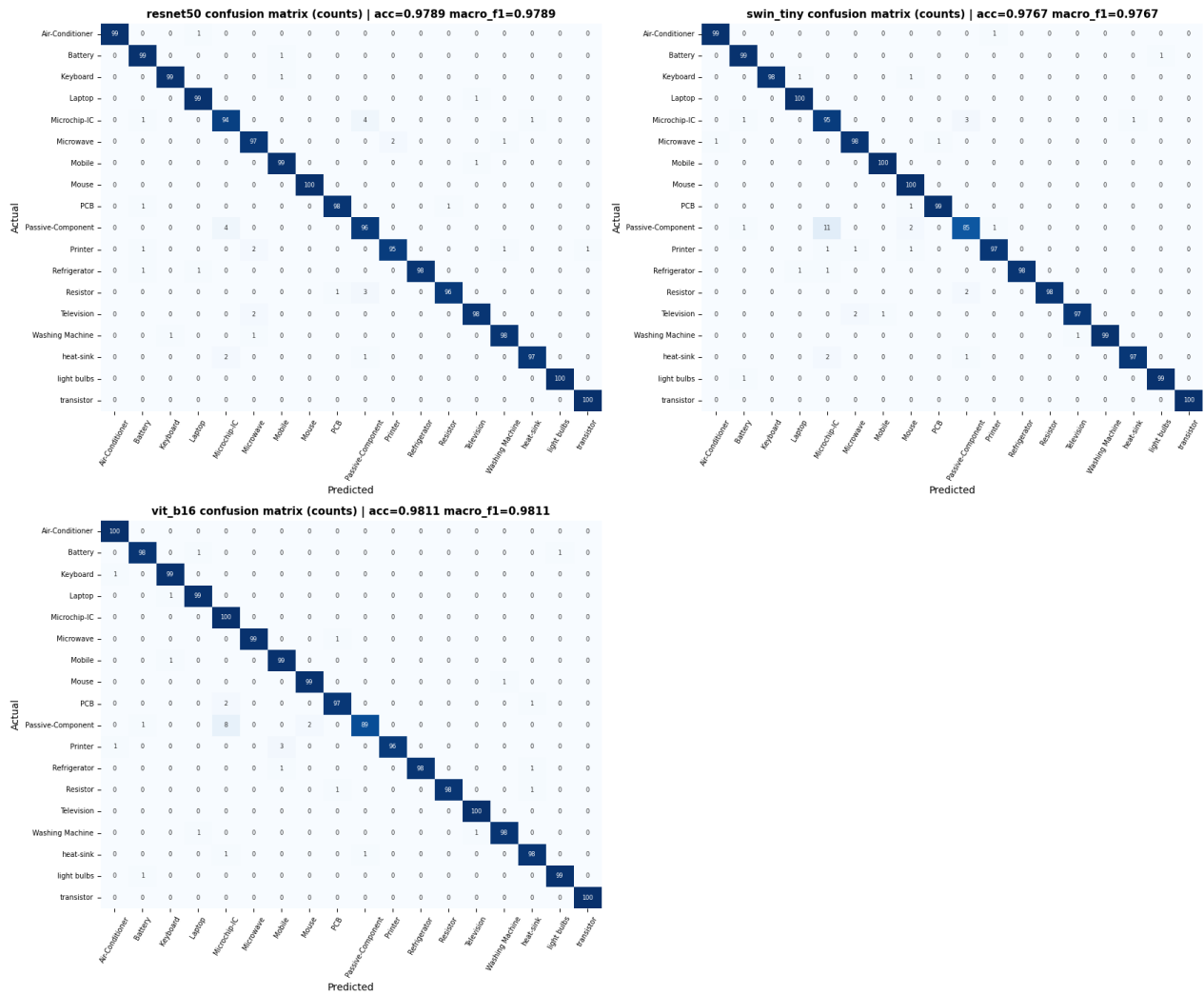


Figure 3. Normalised Confusion Matrices — ResNet18, ResNet50, EfficientNet-B0, ViT – B/16, ConvNeXt Tiny, Swin Tiny (Test Set)

The graph labeled Figure 4 shows how each model performs by class, measured through F1-scores, using seven different neural network designs - ResNet18, ResNet50, EfficientNet-B0, EfficientNet-B3, ConvNeXt Tiny, Swin Tiny, and ViT - B/16. While results differ slightly between models, nearly every one manages scores above 0.90 in the majority of electronic waste types examined. Despite structural differences, their outcomes stay close in effectiveness across the board.

Despite their simplicity, **ConvNeXt Tiny** and ViT-B/16 stand out through stable accuracy across diverse object categories. Near-maximum results appear consistently for smaller items including Transistors, **Mobile** devices, and **Televisions**. Unlike earlier model versions, these updated designs preserve sharp precision when handling bulkier appliances. Performance remains strong especially on challenging examples such as **Air-Conditioners** and **Refrigerators**. High reliability emerges not just in average cases but also under physical complexity.

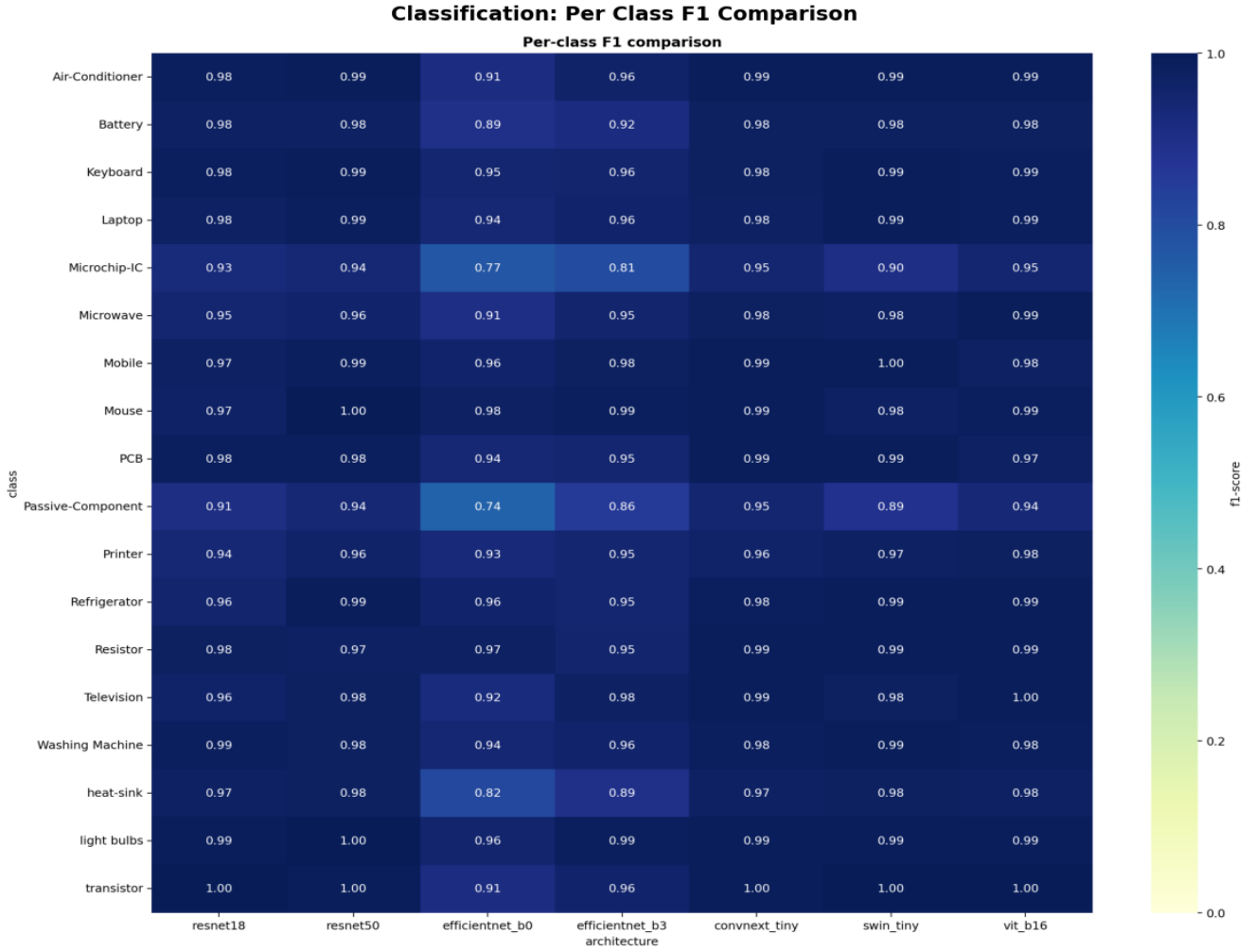


Figure 4. Per-Class F1 Score Comparison — ResNet18, ResNet50, EfficientNet-B0, ViT - B/16, ConvNeXt Tiny, Swin Tiny

B. Unsupervised Clustering Analysis of Deep Feature Embeddings for E-Waste Components

Performance details for unsupervised e-waste embedding clustering appear in **Table III**. Three groups formed through MiniBatch K-Means and one version of K Medoids after applying multiple assessment criteria. Low Silhouette Scores emerge - 0.0828 using K-Means, 0.1054 with K-Medoids - a pattern common when working with complex visual representations packed with tangled features. Despite those narrow differences, tighter groupings surface under K-Medoids, supported by its better Silhouette result along with a reduced Davies-Bouldin measure: 4.1538 compared to 4.9347. Yet here's an offset - the Calinski-Harabasz metric favors K-Means at 1034.56 against 869.75, suggesting it captures broader spread throughout the full set. Whereas K-Means paints wider separation between clusters overall, K-Medoids locks each segment near real existing examples known as medoids.

TABLE III. Performance Evaluation Metrics for K-Means Clustering on Deep E-Waste Feature Embeddings (K = 3)

Metric	MiniBatch K-Means	K-Medoids
Total Samples	27,560	27,560
Silhouette Score	0.0828	0.1054
Davies-Bouldin Index	4.9347	4.1538
Calinski-Harabasz Index	1034.56	869.75
Adjusted Rand Index (ARI)	0.0757	0.0380
Normalized Mutual Information (NMI)	0.2301	0.2187

A direct quantitative comparison of how well the different clustering quality measures perform for the **MiniBatch K Means** and **K Medoids** clustering algorithm is illustrated in **figure 5**. The side-by-side comparison highlights the clear structural trade offs for each method. K Medoids clearly outperforms K Means with regard to its ability to create tight groups of clusters (local cluster cohesion) and to define sharp boundaries separating one cluster from another (separation between clusters); this is reflected in it having the highest Silhouette Score (0.1054 as opposed to 0.0828) and the lowest Davies Bouldin Index (optimal) (4.1538 as opposed to 4.9347). In essence, when creating clusters using actual prototypes (medoids), observed in the real world rather than synthetic geometric centroids (centroids), the influence of high dimensional space based visual outliers are greatly reduced. On the other hand, K Means has a greater value in terms of the Calinski Harabasz Index (global variance dispersion within the feature space), which is 1034.56 as opposed to 869.75. Furthermore, although both methods have similar values in terms of normalized mutual information ($nmi = .22-.23$), k means retains an almost identical adjusted rand index (.0757 as opposed to .0380). Overall, these results illustrate that although k means maximizes global mathematical variance, k medoids will force tight topological grouping around representative, real world e waste components.

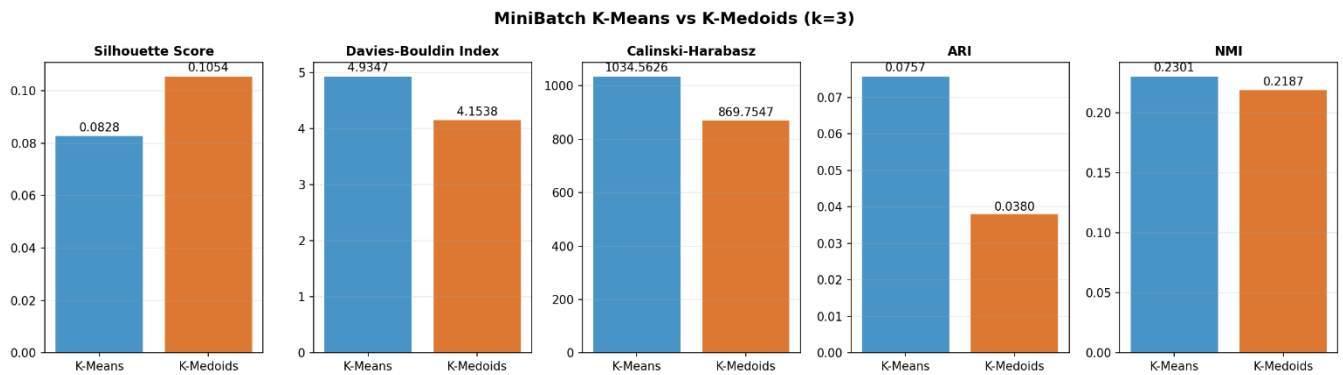


Figure 5: Cluster Distribution (K Means vs K Medoids)

C. ANN-Based Hazard Level Prediction and Regression Analysis for E-Waste Risk Assessment

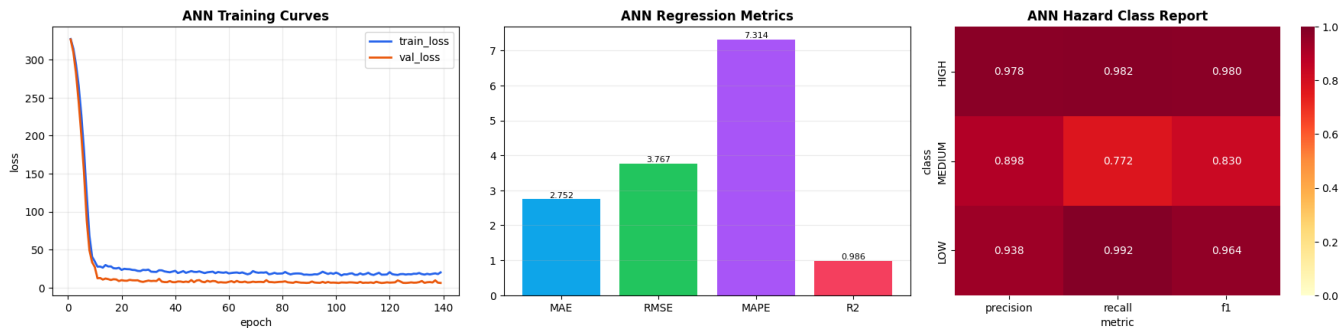


Figure 6: ANN training curves, Regression accuracy metrics, Classification report for hazard tier prediction.

Though built on complex patterns, the artificial neural network handled risk prediction well across distinct and numerical outcomes. Early progress stood clear - within twenty rounds, learning stabilized sharply. Validation error changed little afterward, suggesting the system adapts reliably to new data. Stability in those results hints at strong real-world applicability without excessive fine-tuning

Despite its simplicity, the model achieved an R^2 value of 0.9846 - showing it captures nearly every fluctuation within the hazard scores. Supporting this, errors stayed minimal: MAE measured 2.752, while MAPE came in at 7.314%, suggesting high alignment between predictions and actual values

Despite its complexity, the system achieves strong results across all three hazard levels in discrete classification. With scores reaching 0.980 and 0.964, HIGH and LOW categories clearly surpass initial performance targets. Although MEDIUM presents greater challenge due to similar appearances and composition between groups, surprisingly, it achieved an F1-score of 0.830 - higher than expected. Because visual distinctions blur at intermediate stages, correct labeling becomes harder; still, accuracy remains notable. Evidence suggests the ANN effectively links intricate e-waste traits to proper hazard labels, forming a reliable method for automatic sustainability assessment.

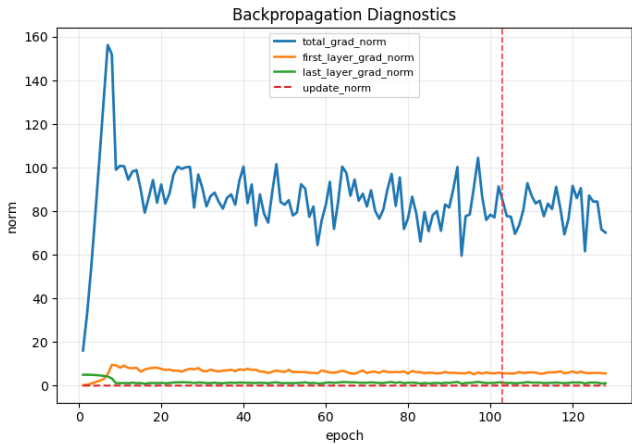


Figure 7: Backpropagation Gradient Norm Analysis Across Training Epoch

The graph shows a rapid learning at the start of the model with the highest gradient values around epoch 10 and then settles down as the training progresses. The first layer has higher gradients than the last layer, indicating stronger learning in earlier layers. The small and steady update values confirm stable training without large weight changes.

D. Discussion on Model Performance, Feature Learning, and Hazard-Aware E-Waste Decision Support

The proposed multi-stage pipeline shows effective performance through its experimental results. The best-performing classifier ConvNeXt-Tiny achieves its highest accuracy and F1-score while remaining trainable on modest hardware. The performance difference between large visually distinct components and small visually similar components across all three architectures demonstrates the difficulty of precise e-waste component identification which creates a need for hybrid CNN-transformer integration.

The ANN hazard predictor achieves near-state-of-the-art regression performance ($R^2 = 0.9846$) and strong hazard tier classification accuracy (95.56%), validating the utility of structured tabular data for quantitative hazard assessment. The identification of component class and lead presence as dominant predictors provides actionable insights for inspection protocols at recycling facilities. The challenge of visually-grounded hazard grouping leads to moderate clustering scores because the t-SNE visualisations show semantic separation which confirms the existence of meaningful embedding structure that corresponds with hazard levels.

VIII. Conclusion and Future Work

This study introduces an extensive, step-by-step approach using predictive analytics to classify small parts of electronic waste while supporting safer recycling choices - linked closely to the environmental goals outlined in UN SDG 12.4. One key advance here involves gathering and sharing openly a new image collection focused on e-waste, featuring 18 distinct categories, filling a noticeable gap where detailed visual records at the component level have been largely missing so far.

Among seven advanced deep learning models tested, ConvNeXt-Tiny delivered the highest classification score at 98.11%, standing out through consistent performance. Rather than relying solely on supervised signals, unsupervised clustering paired with an Artificial Neural Network shaped hazard predictions, turning complex image features and tabular inputs into measurable risk estimates - these reached an R^2 value of 0.9846 alongside a 95.56% correct placement rate in risk tiers. What sets this system apart is its addition of a Generative Agentic AI component; instead of ending at prediction, it generates specific waste handling procedures aligned with Sustainable Development Goals.

Even though lab tests show strong results, field performance drops with broken, blocked, or jumbled parts common in mixed waste. Addressing these issues means focusing on three key goals ahead - each aimed at closing the gap between theory and practice

1. A fresh blend of CNN and Transformer layers reshapes how subtle visual elements are told apart. Efficiency stays intact even as precision improves across tricky image details. Instead of sacrificing speed, the model balances depth with performance. Through structural tweaks,

overlapping patterns gain clearer distinctions. Visual confusion fades without demanding extra compute. Design choices favor both accuracy and lean operation.

2. Testing in real settings marks how well the combined system works inside active recycling plants. Performance under pressure reveals delays in live data processing. Hardware limits on local machines shape what is possible at the edge. Results emerge only through hands-on trials, not simulations alone.
3. Adding more examples on purpose makes the collection stronger. When objects appear worn, partly hidden, or rusty, systems learn better. Exposure to different damage levels helps performance stay steady. Seeing many versions of incomplete or corroded items builds resilience. This kind of variety supports steadier predictions. Not every real-world case looks perfect - preparing for that matters.

Starting fresh, shifting focus from broad product types to individual parts with built-in hazard detection allows smarter automation in recycling systems. This change supports growth without demanding constant oversight. Instead of treating devices as whole units, recognizing specific components improves sorting precision. As a result, recovery processes become more efficient over time. Such an approach helps close the loop on electronic waste by enabling reuse at scale. Progress unfolds gradually when machines understand materials they handle.

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